Games — problem 13.25

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Let n be a positive integer and $M = \{1, 2, 3, 4, 5, 6\}$. A starts with any digit from M. Then B appends to it a digit from M, and so on, until they get a number with $2n$ digits. If the result is a multiple of 9, then A wins ; otherwise B wins. Who wins, depending on n ?